

Practical - 3

Practical Title: Responsive UI Design using CSS Grid, Flexbox, and Bootstrap/Tailwind

Problem Definition: Design responsive layouts for Home, About, Registration, Dashboard, and Events pages using CSS Grid and Flexbox. Optional to use CSS framework such as Bootstrap or Tailwind. Ensure mobile-first responsiveness.

F. Theory

1. How does the layout adapt for mobile, tablet, and desktop screens?

Answer: A layout adapts across mobile, tablet, and desktop screens using fluid grids, flexible images, and breakpoints (typically under 600px for mobile, 601px–1024px for tablets, and 1025px+ for desktops). Elements reflow, resize, or stack vertically to fit the available screen space.

(https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Responsive_Design)

2. Which layout technique is used and why?

Answer: Common web design layout techniques include **CSS Flexbox**, **CSS Grid**, and traditional **Float** or **Responsive** structures. Because your specific project or graphic was not provided, the correct technique depends entirely on the design goal.

(https://www.w3schools.com/html/html_layout.asp)

3. Are typography, spacing, colors, and contrast readable?

Answer: Yes, typography, spacing, colors, and contrast directly determine whether text is readable. When properly balanced, they prevent eye strain and make content easy to scan. When poorly configured, text becomes difficult or impossible to read.

(<https://guidelines.india.gov.in/activity/the-impact-of-typography-on-web-design-a-comprehensive-guide-to-aesthetics-and-readability/>)

4. Is the framework used consistently without unnecessary inline CSS?

Answer: `style=""` attributes. While they look like inline styles in the markup, they compile into clean external stylesheets and maintain design consistency without unnecessary true inline CSS.

(https://www.reddit.com/r/Frontend/comments/rk3lr4/are_css_frameworks_really_necessary/)

G. Implementation/ Output/Result :

- CSS Selectors Used:

*, body, header, nav, main, section, article, footer, div,
h1, h2, h3, h4, p, ul, li, a, img,
form, label, input, textarea, select, button,
table, tr, td, hr, small, i, marquee,
.class, #id,
:hover, :focus, :first-child, :last-child,
input[type="text"], input[type="email"], input[type="password"],
input[type="radio"], input[type="checkbox"],
input[type="submit"], input[type="reset"],
@media

- CSS Properties Used:

margin, padding, box-sizing,
display, width, height, max-width,
background, background-size, background-position, background-repeat,
color,
font-family, font-size, font-weight,
text-align, text-decoration, line-height,
border, border-left, border-top, border-radius,
box-shadow,
position,
justify-content, align-items,

flex-direction, flex-wrap,

gap,

grid-template-columns,

list-style,

cursor,

transition,

transform,

outline,

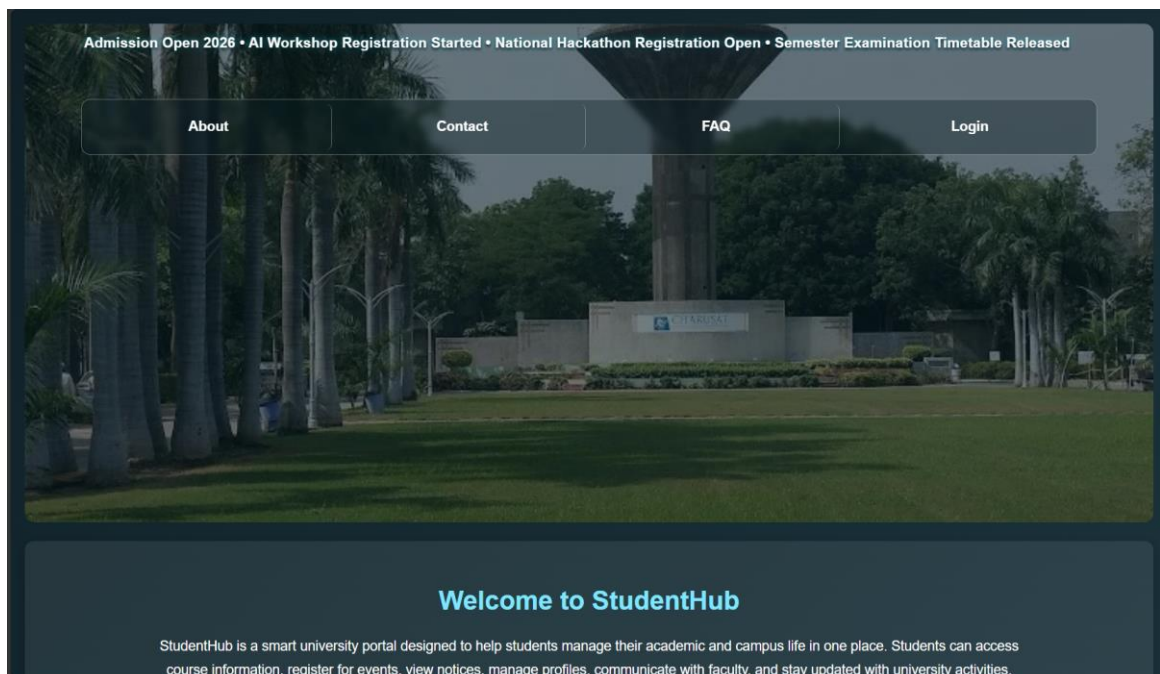
resize,

object-fit,

overflow

H. Evidence of Work (Screenshots)

Figure 3.1(Home page)



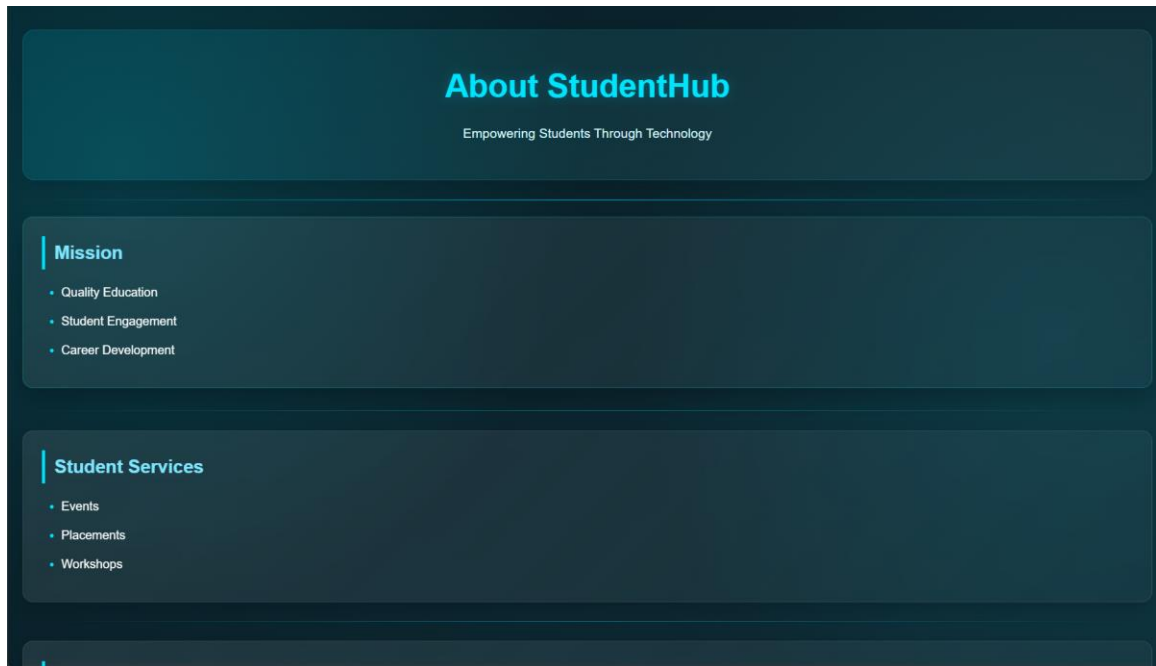


Figure 3.2(About page)

The screenshot shows the 'StudentHub Portal' registration page. At the top, the CHARUSAT logo is displayed, followed by the title 'StudentHub Portal' in a large, bold, light blue font. Below the title, the text 'Create Your Account' is displayed, followed by a link 'Already have an account? Login Here'. The registration form is titled 'Registration Form' and contains the following fields: 'Full Name' (text input), 'Email' (text input), 'Mobile' (text input), 'Password' (text input), 'Confirm Password' (text input), 'Date of Birth' (text input with a date picker icon), and 'Gender' (radio buttons for Male, Female, and Other). The background is a dark teal color with a subtle grid pattern.

Figure 3.3(Registration page)

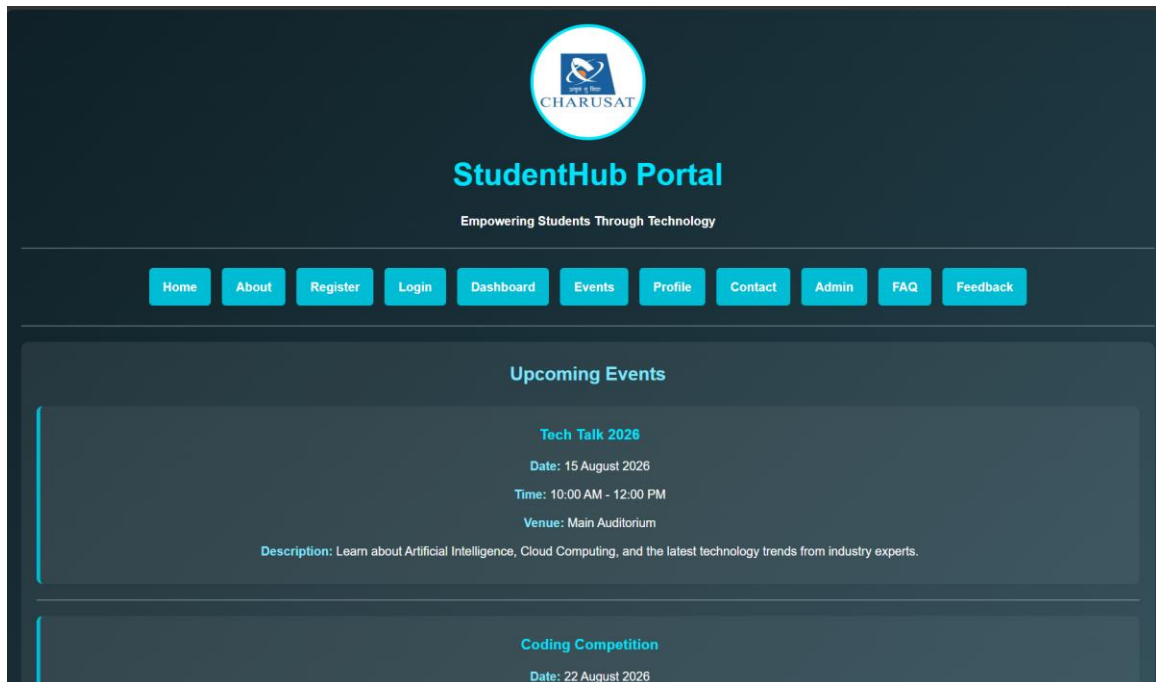
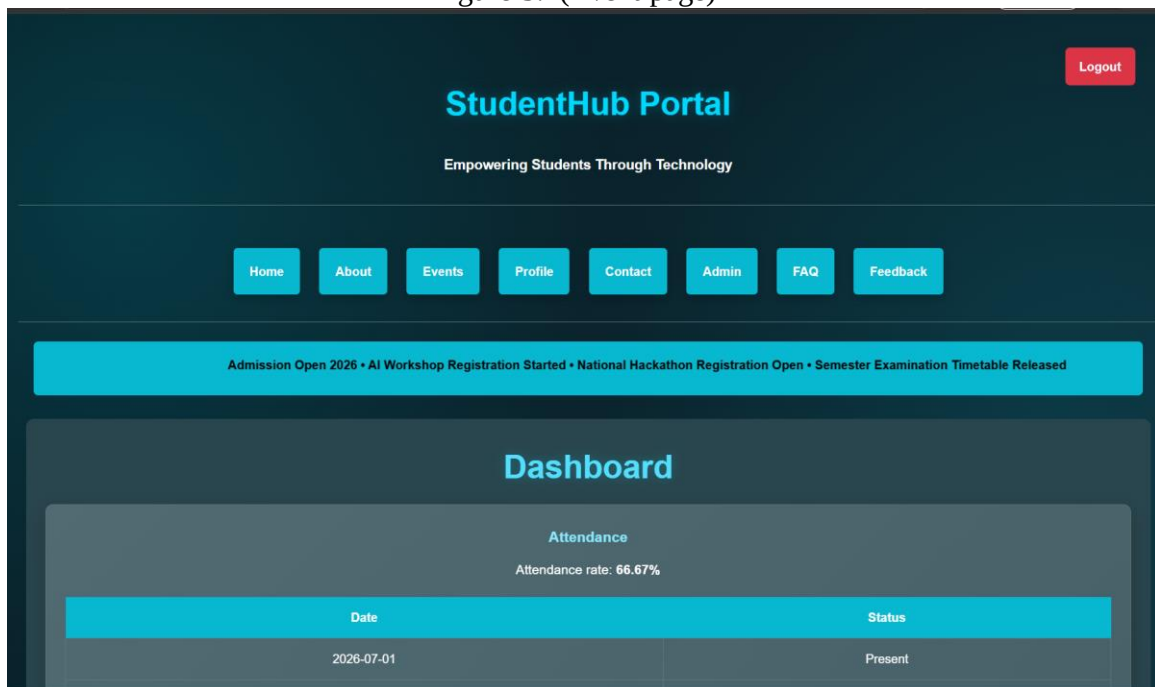


Figure 3.4(Event page)



Attendance rate: 66.67%		
Date		Status
2026-07-01		Present
2026-07-02		Absent
2026-07-03		Present

Timetable		
Day	9:00-10:00	10:15-11:15
Mon	Mathematics	Physics
Tue	English	Computer Science
Wed	Chemistry	Lab

Results	
Course	Grade
Mathematics	A
Physics	B

Figure 3.5(Dashboard page)



StudentHub Portal

Empowering Students Through Technology

[Home](#)
[About](#)
[Register](#)
[Login](#)
[Dashboard](#)
[Events](#)
[Profile](#)
[Contact](#)
[Admin](#)
[FAQ](#)
[Feedback](#)

Student Hub Feedback Form

Thank you for using Student Hub. Your valuable feedback helps us improve our platform and provide a better learning experience.

Personal Information

Full Name :

Email Address :

Figure 3.6 (Feedback Form)

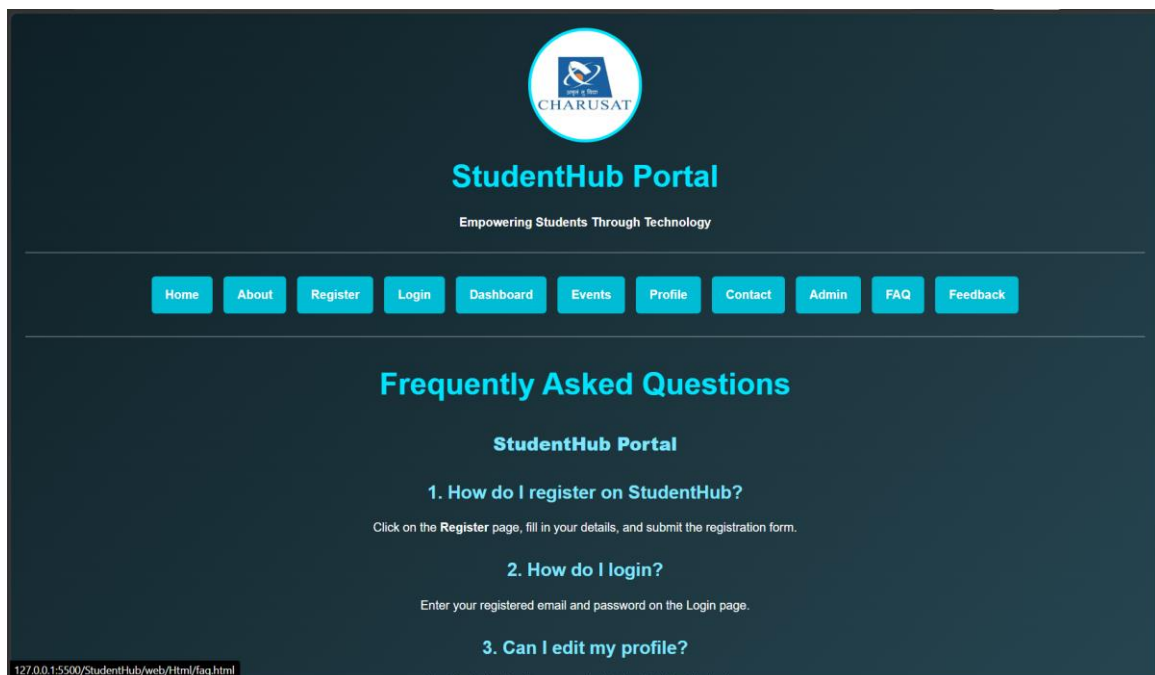


Figure 3.7 (Faqs)

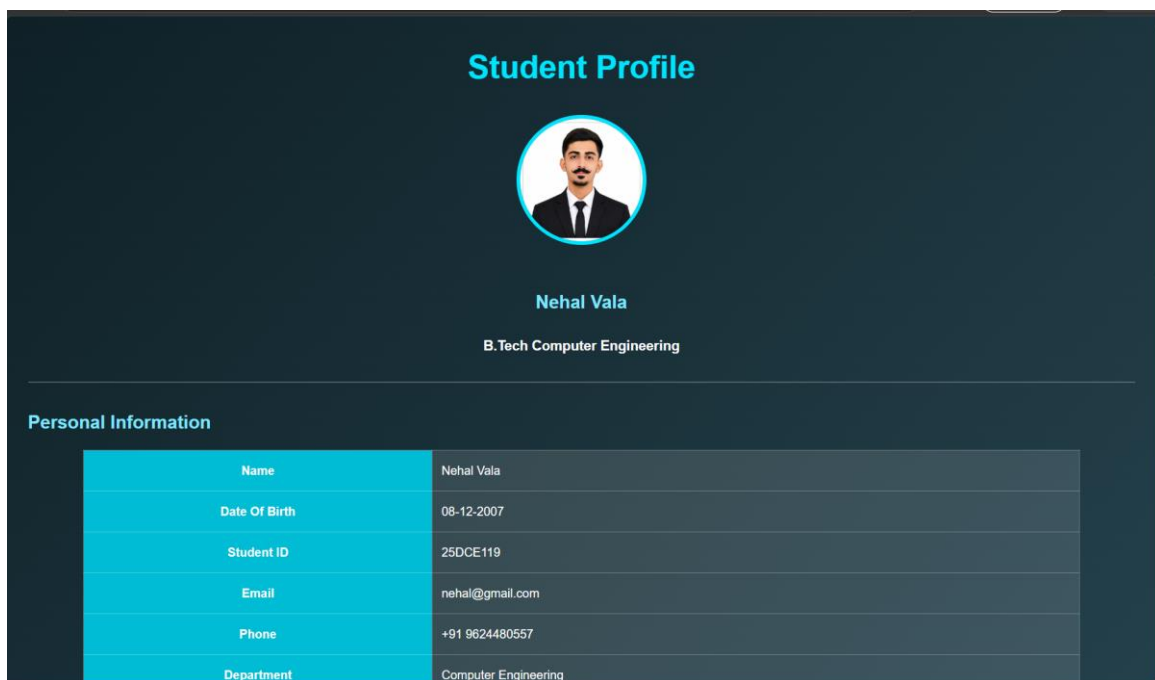


Figure 3.8 (Profile)

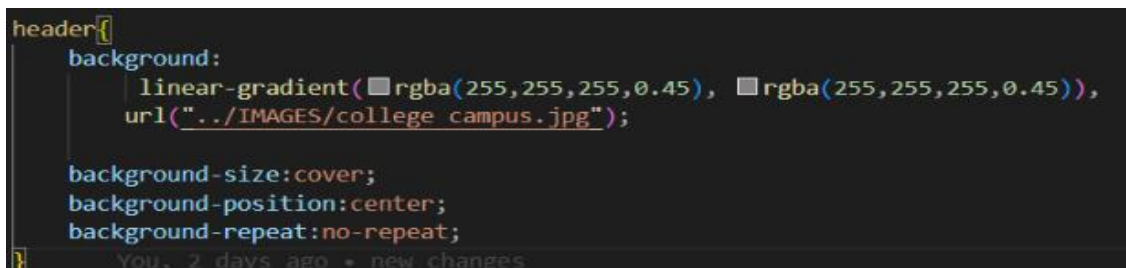
I. GitHub Repository Link:

<https://github.com/nikhilvala0005-cloud/sh.git>

J. Challenges Faced:

When I was styling the html files , I had no idea about how I put image behind the text and blur it . then I told ChatGPT that I want to put image behind text at header which concept or styling function I should use , give syntax and opacity code and gradient code like (255,255,255,0.20);

It gave me function that name is linear gradient and also I learned padding , card , form, Flex , grid, styling , box , responsive etc.. concept.



```
header{
  background:
    linear-gradient(rgba(255,255,255,0.45), rgba(255,255,255,0.45)),
    url("../IMAGES/college campus.jpg");
  background-size:cover;
  background-position:center;
  background-repeat:no-repeat;
}
```

K. Learning Outcomes:

I will design a professional, responsive, and reusable UI layout.

Conclusion :

The StudentHub Practical successfully demonstrated the design and development of responsive user interfaces using CSS Grid, Flexbox, and responsive design techniques. The Home, About, Registration, Dashboard, and Events pages were styled with proper layouts, spacing, typography, colors, cards, forms, and navigation elements to provide a consistent user experience across different screen sizes.

Through this practical, I learned how to create flexible and responsive layouts using CSS properties such as Flexbox, Grid, media queries, gradients, padding, margins, and other styling techniques. I also learned how to adapt web pages for mobile, tablet, and desktop screens and improve the overall visual appearance of the StudentHub portal. The challenges faced while implementing background images, opacity, gradients, and responsive

components helped strengthen my practical understanding of CSS.

Overall, this practical improved my ability to design professional, responsive, and reusable web interfaces and provided a strong foundation for developing modern and user-friendly websites.

Faculty Remarks: (Keep this blank for faculty)